

Kalshi Prediction Market Model

Week 3 Research Update · Damien Nguyen · CSC 4999 · Advisor: Prof. Keith Mills

v2 results: current dataset

Change to the standard. I propose replacing the +0.005 macro F1 keep rule with a significance test on log loss: a feature is kept if adding it lowers log loss with a one-sided $p < 0.05$ on a Diebold–Mariano test (details in the significance standard section below). Macro F1 is still reported. The tables therefore show log loss alongside macro F1.

Macro F1 from walk-forward CV on the current (post-Spring) development events, with 5 training seeds. Majority and no change both always predict FLAT, so they share a row. Log loss (lower is better) is the mean $\pm$ sd over the same 5 seeds. Baselines are scored on probabilistic forms and have no seeds, so no sd: majority, no change and random use the training fold's class rates, and momentum uses the training rates of each label given the market's previous label (smoothed toward the class rates).

All events (10,371)

Predictor	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Mean $\pm$ sd	Log loss (mean $\pm$ sd)
Majority / no change	0.307	0.307	0.307	0.307	0.307	0.307	0.531
Random	0.336	0.334	0.330	0.331	0.325	0.331 $\pm$ 0.004	0.531
Momentum	0.461	0.461	0.461	0.461	0.461	0.461	0.462
Price	0.504	0.501	0.505	0.501	0.503	0.503 $\pm$ 0.002	0.606 $\pm$ 0.003
Price + sentiment	0.513	0.515	0.513	0.518	0.519	0.516 $\pm$ 0.003	0.587 $\pm$ 0.003

News events: ≥ 2 articles in the 6 hours before prediction time (4,559)

Predictor	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Mean $\pm$ sd	Log loss (mean $\pm$ sd)
Majority / no change	0.280	0.280	0.280	0.280	0.280	0.280	0.786
Random	0.342	0.330	0.335	0.331	0.324	0.332 $\pm$ 0.007	0.786
Momentum	0.451	0.451	0.451	0.451	0.451	0.451	0.707
Sentiment alone	0.390	0.397	0.400	0.406	0.401	0.399 $\pm$ 0.006	0.927 $\pm$ 0.001
Price	0.514	0.520	0.519	0.514	0.521	0.518 $\pm$ 0.003	0.703 $\pm$ 0.002
Price + sentiment	0.541	0.544	0.536	0.536	0.535	0.538 $\pm$ 0.004	0.673 $\pm$ 0.001

Finding: price + sentiment differs from price on few events

On the 4,559 news events, the two models make the same call on 86.5%. Where they differ, price + sentiment is right more often: 261 events against 173, a net of +88 correct calls (1.9% of events). The probability on the true class moves by about 0.16 either way, so the fixes and breaks are equally confident. The full breakdown is at the bottom of the doc.

Finding: the matched articles are mostly unrelated to the market

Measure	Value
Articles per news event	Mean 14.4, median 12
Articles containing the market's exact query phrase	3.4% (0.50 per event)
Articles with any query word in the headline	5.5% (0.80 per event)
Events with ≥ 2 articles containing the query phrase	233 of 4,559 (5.1%)
Events with ≥ 2 articles with a query word in the headline	865 of 4,559 (19.0%)
Article overlap between unrelated markets in the same hour	Mean Jaccard 0.32

The search queries are built from overlapping word pairs in the market title, such as high temp temp Miami Miami Nov, and the search returns general news for them.

Proposed fix, before the search arms start

- 1 Rebuild the queries from the market's topic and keep only articles that mention it.
- 2 Add hour, weekday and article volume to the price baseline, so sentiment has to beat them.
- 3 Use different-market sentiment as the placebo feature in the standard below.
- 4 Re-run price + sentiment vs price.

A strict relevance filter leaves 233 to 865 news events at current coverage, so step 1 likely also needs a longer lookback than 6 hours or a wider news source.

Proposed next step: significance standard

I propose replacing the +0.005 macro F1 keep rule with a significance test: a feature is kept if adding it lowers log loss with a one-sided $p < 0.05$ on a Diebold–Mariano test.

Diebold–Mariano test. The standard test for whether two forecasters differ in accuracy. It takes the difference in loss between them on each observation and tests whether the mean difference is zero, using a standard error that allows the differences to be correlated over time.

Event group. All the markets Kalshi lists under one event: the same question at different strikes, such as every "S&P 500 closes between X and Y on Feb 23" market. Their prices move together, so each event group counts as one observation, not each market.

Step	
1. Predict	Average each event's probabilities over 5 training seeds, with and without the feature
2. Difference	$d = \text{log loss}(\text{without}) - \text{log loss}(\text{with})$ per event, summed within each event group
3. Test	Diebold–Mariano on the event-group sums in time order; keep if one-sided $p < 0.05$
4. During search	The gain must also beat the 95th percentile of placebo features, with Benjamini–Hochberg correction across candidates
5. Final claim	LLM vs TPE and vs random on the sealed slice, scored once, Holm-corrected

Macro F1 is still reported, with an event-group bootstrap p-value.

Dropped

Dropped	Why
+0.005 macro F1 threshold	Below the noise: CIs are about ± 0.011 , so it never decides the outcome
At least 4 of 5 folds won	Too coarse with 5 folds; can reject a feature whose CI excludes zero
Single training seed	Verdict flips by seed: sentiment vs price is significant on 3 of 5 seeds
Macro F1 as the deciding metric	Ignores predicted probabilities and is weak when most events are FLAT
Bootstrap CI as the deciding test	Treats event groups as independent; groups open at the same time are correlated, so its p-values are too small

Next moves

- 1 Save each event's predicted probabilities from CV (only scores are saved today).
- 2 Implement the test and the placebo threshold.
- 3 Re-run price + sentiment vs price under the new standard.
- 4 Use it as the keep rule for the search arms.

Additional stats: price vs price + sentiment, where they disagree

News events on the current dataset (4,559), predictions averaged over the 5 seeds. The two models make the same call on 86.5% of events.

Outcome	Events
Both right	2,994
Both wrong	1,131
Only price + sentiment right	261
Only price right	173

Sentiment nets +88 correct calls. When it changes the outcome, the probability on the true class moves by about 0.16 either way.

Sentiment helped (261 events, price wrong and price + sentiment right):

What price got wrong	Events
Predicted a move, but the market stayed FLAT	99
Predicted the wrong direction (UP vs DOWN)	151
Predicted FLAT, but the market moved	11

Sentiment hurt (173 events, price right and price + sentiment wrong):

What price + sentiment got wrong	Events
Predicted a move, but the market stayed FLAT	38
Predicted the wrong direction (UP vs DOWN)	115
Predicted FLAT, but the market moved	20

Where the code is

Repository: <https://github.com/damienwin/kalshi-search-efficiency>

The feature search for Workstream C is built and tested; no search has been run yet.

Part	Status
Feature grammar	Candidate features are expression trees over 28 operators and the 14 price columns; no generated code runs, and every operator is tested for look-ahead
Search arms	Random, TPE, and LLM full / no-history / blind (C1–C3), drawing from the same grammar under the same trial budget
Significance gate	The test above, with a variance that also allows for correlation between markets open on the same days
Search loop	Logs every trial; LLM calls are recorded so a run can be replayed without API access

Part	Status
C4	LLM prompts contain no dates, tickers or event names; gains can be split before and after the model's training cutoff
Tests	561 pass, including end-to-end runs of all five arms on synthetic data

Next: one live test of the LLM arm, then 5 seeds per arm.